

EVIDENCE REPORT | STATIONS 3–4

MarketLens

Walk-forward funds, finance-aware sentiment, and a governed fusion experiment

FINS5545 Assignment 3 / Project B

12	36	146,836	2021–23
OOS FUNDS	MONTHLY DECISIONS	HEADLINES	LIVE SAMPLE

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Course data through 31 December 2023 • Prepared August 2026

Report stance

Historical evidence is reported after a 10 bps one-way cost unless stated. The sentiment candidate is not promoted as alpha because its discovery-window selection is zero and its small holdout benefit is cost-fragile.

01

Executive evidence

MarketLens is intended for investors who want to compare risk and return in a clear and understandable way.

MarketLens uses the cleaned Project A price and news data to construct equity, crypto, and combined research portfolios. Four portfolio rules are applied to each universe. All reported returns use weights estimated from information available before the relevant investment period. In the app, users can compare the twelve funds, review an individual fact sheet, examine news sentiment, and test an illustrative allocation after fees.

12.6%	0.82	-22.4%	0.95
EQUITY EW RETURN	EQUITY EW SHARPE	COMBINED RP MDD	CRYPTO MV SHARPE

What the evidence says

Among the equity funds, Equal-Weight has the highest net Sharpe ratio (0.82), with a 12.6% annualised return and 16.1% volatility. This also provides a useful check on estimated-return optimisation: Maximum-Sharpe trades more frequently but does not outperform the simpler rule.

Crypto Minimum-Variance records the highest annualised return and Sharpe ratio in this sample. However, its 74.3% volatility and -72.2% maximum drawdown indicate that it is better viewed as a high-risk satellite allocation than as a general default.

Combined Risk Parity has a more moderate mixed-asset profile: a 12.7% annualised return, 17.4% volatility, a 0.77 Sharpe ratio, and a -22.4% maximum drawdown. Combined Equal-Weight earns more, but at the cost of higher volatility and a larger drawdown.

Innovation conclusion

The pre-set +0.25 sentiment tilt raises 2023 net Sharpe from 0.51 to 0.52 at 10 bps, but 2021–2022 discovery selects $\lambda=+0.00$. The small 2023 advantage disappears near 25 bps. Sentiment remains research analytics pending stronger evidence.

02

Backtest and fund architecture

Following the Week 10 framework, the backtest uses expanding estimates, monthly walk-forward decisions, and the appropriate calendar for each asset class.

Past-only decision rule

Calendar year 2020 forms the initial estimation window. From 2021, each portfolio is rebalanced on the first observed date of the month. At every decision date, weights are estimated only from earlier observations and are then held until the next rebalance, with holdings allowed to drift between decisions. Initial turnover is set to 100%. Thereafter, turnover is half the L1 distance between drifted pre-trade holdings and the new target weights. One-way trading costs equal turnover multiplied by the assumed cost rate.

Family	Assets	Calendar	Annualise	First live	Cap
Equity	50	Equity trading days	252	2021-01-04	20.0%
Crypto	10	Calendar days	365	2021-01-01	25.0%
Combined	60	Equity trading days	252	2021-01-04	20.0%

Table 1. Walk-forward contract by asset family.

Source: results/tables/backtest_design.csv; student calculations.

Four methods, one controlled comparison

Equal-Weight allocates $1/N$ to each asset and estimates no moments. Minimum-Variance uses the expanding covariance matrix to minimise portfolio risk. Maximum-Sharpe also uses expanding mean returns and is therefore more exposed to estimation error. Risk Parity seeks to equalise volatility contributions. All portfolios are long-only and fully invested. Individual weights are capped at 20% in the equity and combined universes and 25% in crypto.

Calendar and annualisation choices

Simple returns for equities and crypto are calculated separately on their native calendars. For the combined universe, crypto returns are then sampled on the equity calendar. This avoids creating an artificial one-day return from a weekend price gap after merging price levels. Equity and combined portfolios use 252 annual periods, while crypto uses 365. Sharpe ratios assume a zero risk-free rate, consistent with the course examples.

Automated audit

A synthetic test replaces all future returns after a cutoff and verifies that earlier monthly weights are unchanged. Generated target weights also pass sum-to-one, non-negativity, and cap checks.

03

Fund evidence and customer choice

There is no single best fund on every measure. The preferred choice depends on how much the user values growth, protection from large losses, and simplicity.

Fund	Return	Vol.	Sharpe	MDD	Turnover/yr
Equity Equal-Weight	12.6%	16.1%	0.82	-20.3%	0.65x
Combined Risk Parity	12.7%	17.4%	0.77	-22.4%	0.71x
Crypto Minimum-Variance	53.6%	74.3%	0.95	-72.2%	1.13x
Combined Maximum-Sharpe	7.1%	32.7%	0.37	-48.6%	2.08x

Table 2. Selected net fact-sheet metrics, monthly OOS, 2021–2023.

Source: results/tables/performance_metrics.csv. Return is CAGR; cost is 10 bps one-way; zero risk-free rate.

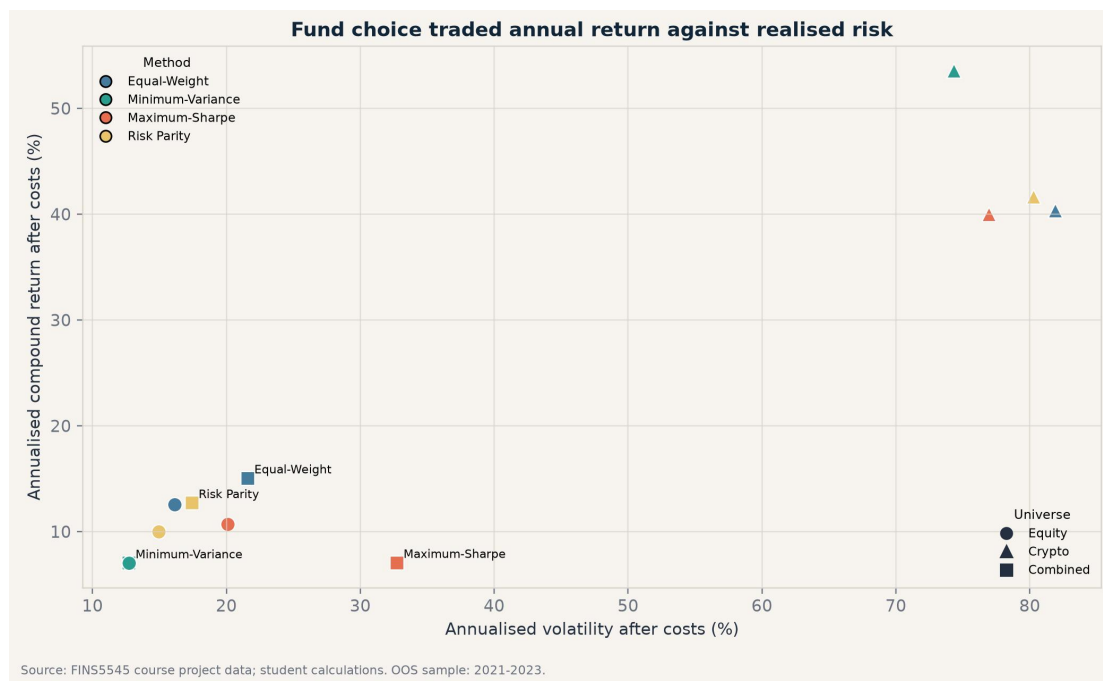


Figure 1. Net annualised return versus realised volatility for all twelve funds, 2021–2023.

Source: FIN5545 prices; student calculations. Equity/combined annualised at 252, crypto at 365; 10 bps one-way cost.

Crypto funds appear on a substantially different risk scale from the other universes. Within equities, Equal-Weight achieves the highest return without having the highest volatility. The weak performance of Combined Maximum-Sharpe illustrates the practical cost of estimation error.

Interpretation boundary

These rankings describe one short regime. They do not establish expected future dominance. The app therefore shows the full distribution of trade-offs rather than presenting a single ‘best fund’ badge.

04

Fact sheets, holdings, and risk

For each of the twelve funds, the app shows growth, drawdown, performance measures, turnover, and the most recent target allocation.

Combined method	Return	Sharpe	MDD	Top current targets
Equal-Weight	15.0%	0.76	-27.9%	ABBV 1.7%; ABT 1.7%; ADA-USD 1.7%
Minimum-Variance	7.0%	0.60	-17.8%	WMT 19.9%; MRK 12.8%; ABBV 11.9%
Maximum-Sharpe	7.1%	0.37	-48.6%	ABBV 20.0%; NVDA 20.0%; NUE 20.0%
Risk Parity	12.7%	0.77	-22.4%	WMT 3.2%; GILD 3.1%; MRK 3.0%

Table 3. Combined-family fact-sheet snapshot and top three December 2023 targets.

Source: results/tables/performance_metrics.csv and results/data/current_holdings.csv. Top holdings are target weights, not drifted end-of-day weights.

What changed across methods

At each rebalance, Equal-Weight allocates 1.67% to every combined-universe asset, creating a 16.7% crypto sleeve. Minimum-Variance assigns almost no weight to crypto because crypto risk dominates the covariance objective; in practice, its results are therefore close to Equity Minimum-Variance. Risk Parity adjusts more gradually while maintaining broader diversification. Maximum-Sharpe makes larger changes to sector and crypto weights as estimated returns move.

Investor-facing risk language

Returns should always be considered alongside maximum drawdown. Even the best-performing crypto fund lost more than 70% from peak to trough. The combined portfolios also experienced substantial losses: Combined Equal-Weight fell 27.9%, compared with 22.4% for Combined Risk Parity. Historical Sharpe ratios are therefore presented as descriptions of the sample, not as predictions or guarantees.

Recommended default journey

Open on Equity Equal-Weight as the transparent benchmark; show Combined Risk Parity as a diversified alternative; place crypto funds behind a high-volatility warning. Let users compare rather than preselecting based on the highest historical return.

05

Finance-aware sentiment index

The sentiment index summarises headline tone using transparent aggregation rules and a clearly specified lag for live use.

Headline titles are scored once with plain VADER and once with finVADER, with casing, punctuation, and negation retained. finVADER extends VADER using the SentiBigNomics and Henry finance lexicons (Consoli et al., 2022; Henry, 2008; Korab, 2023). Scores are first averaged within each ticker-day. Sector values then assign equal weight to tickers so that firms with heavier news coverage do not dominate the index.

Model	Headline rows	Mean	Exactly neutral	Positive	Negative
Plain VADER	146,830	0.106	48.8%	37.9%	13.2%
finVADER	146,830	0.069	16.9%	64.5%	18.6%

Table 4. Descriptive sentiment-model coverage on the course headline panel, 2020–2023.

Source: results/tables/sentiment_model_validation.csv. Shares classify compound score relative to zero; repeated news rows remain part of ticker-day coverage after exact duplicate removal.

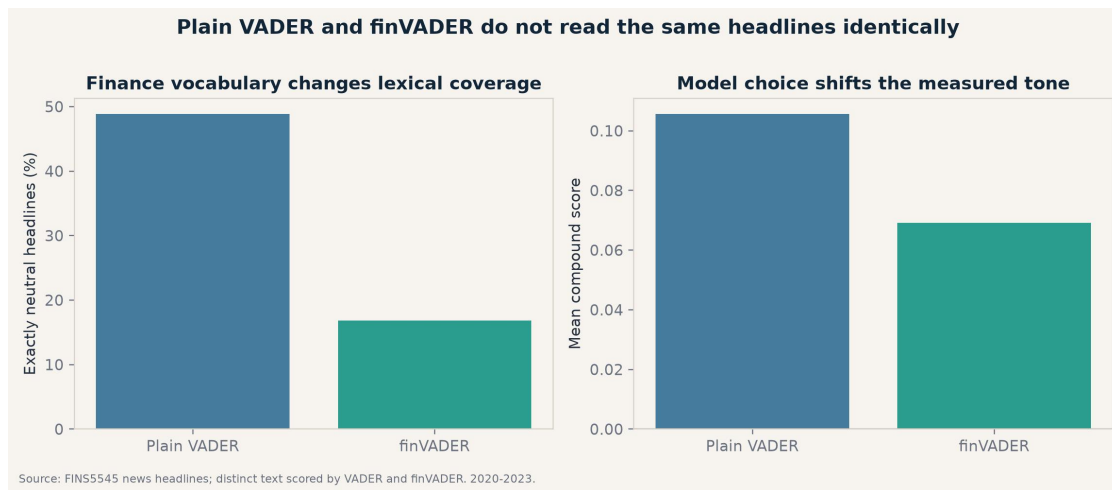


Figure 2. Plain VADER versus finVADER classification coverage and mean tone.

Source: FIN5545 news headlines; student calculations; 2020–2023.

finVADER reduces the proportion of exactly neutral classifications from 48.8% to 16.9%, suggesting that finance-specific words carry information that plain VADER misses. Its lower mean score also shows that model choice materially affects the index. This comparison remains descriptive; predictive value would require a separate test.

Missing days and live use

A ticker's last score carries for at most five equity trading days, then becomes neutral. The filled signal is shifted one full trading day before 21-day smoothing. Fear/greed standardisation uses expanding history; the full-sample z-score is descriptive only.

06

Innovation: fusion with an honest null

The innovation test introduces finance-specific sentiment into the allocation process and then evaluates whether any improvement survives higher turnover and transaction costs.

At each monthly equity rebalance, the lagged 21-day finVADER signal is standardised across stocks and winsorised at ± 2.5 . Equity Minimum-Variance target weights are multiplied by $\exp(\lambda z)$ and then normalised, preserving long-only full investment. A modest positive tilt, $\lambda = +0.25$, is specified in advance for the required before-and-after test. A separate grid from -0.75 to $+0.75$ is evaluated only during 2021-2022, leaving 2023 as an untouched holdout.

2023 holdout	Return	Vol.	Sharpe	MDD	Turnover
Base Equity Minimum-Variance	5.1%	11.0%	0.51	-8.6%	0.27x
Finance-Sentiment Tilt	5.3%	11.0%	0.52	-8.5%	1.04x

Table 5. Required before/after fusion test on the untouched 2023 holdout, net of 10 bps one-way costs.

Source: results/tables/fusion_performance.csv. Base is Equity Equal-Weight; signal is lagged finVADER.

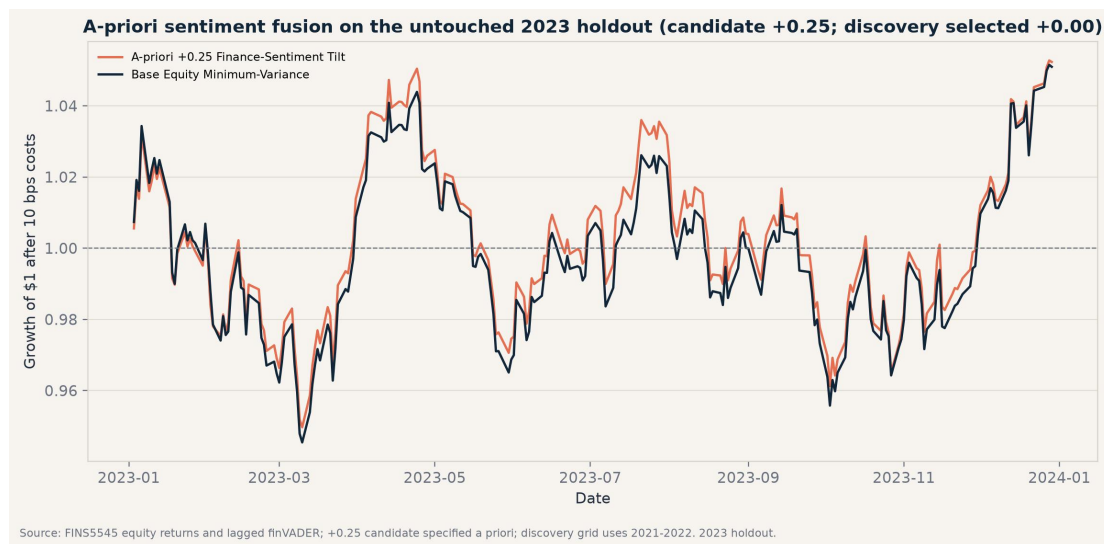


Figure 3. Growth of \$1 for the base and pre-set sentiment candidate in 2023.

Source: FIN5545 equity returns and headlines; student calculations; 10 bps one-way cost.

The pre-specified candidate finishes slightly ahead in 2023, but the margin is narrow and sensitive to transaction costs. The discovery-period grid selects $\lambda = 0.00$, while the $+0.25$ candidate creates more turnover. The evidence therefore does not support treating the tilt as a reliable source of alpha.

Governance decision

Do not productise the tilt. Preserve it as a monitored research layer, require rolling pseudo-holdouts across additional regimes, and introduce an explicit turnover budget before reconsideration.

07

Investor journey and recommendations

The app organises the evidence around five tasks that users can complete without rerunning the underlying models.

App task	What the user can do
Compare	Filter universes; rank all twelve funds; overlay growth paths.
Fact sheet	Choose any fund; inspect growth, drawdown, net metrics, turnover, and latest holdings.
Allocation lab	Blend 2–4 funds with normalised sliders and an annual fee; view an illustrative constant-mix path.
Sentiment	Read fear/greed, sector tone, model coverage, fusion holdout, and cost scenarios.
Method & limits	See the backtest contract, no-look-ahead controls, and non-advice warnings.

Table 6. MarketLens precomputed Streamlit customer journey.

Source: streamlit_app.py. The deployed app reads committed results CSVs only; it performs no NLP, download, or optimisation.

Recommendation 1 — launch comparison before recommendation

Use Equity Equal-Weight as the transparent equity benchmark and Combined Risk Parity as the mixed-asset alternative. For each fund, show return together with volatility, maximum drawdown, turnover, and holdings.

Recommendation 2 — treat crypto as a risk-labelled satellite

The strong historical crypto return should not obscure drawdowns above 70%. For a general-investor interface, crypto allocation controls should therefore be accompanied by a clear high-risk acknowledgement.

Recommendation 3 — research sentiment under a turnover budget

Keep the sector and fear/greed indices as explanatory analytics. Future fusion tests should use rolling discovery and holdout windows, an explicit monthly turnover limit, several pre-specified smoothing settings, and cost thresholds chosen before the results are known.

Critical reflection

The most valuable innovation finding is not the small positive 2023 difference; it is the process that prevents that difference from becoming a claim. Model coverage, timing, selection, turnover and cost all have to survive separate checks.

The main discussion ends here (eight pages including the cover). The appendices contain the required exhibits, detailed results, reproducibility checks, and references; they are excluded from the narrative page limit.

APPENDIX A

Required exhibit: full performance table

All twelve funds; monthly walk-forward OOS; 2021–2023; net of 10 bps one-way costs.

Fund	Return	Vol.	Sharpe	MDD	Turnover/yr
Equity Equal-Weight	12.6%	16.1%	0.82	-20.3%	0.65x
Equity Maximum-Sharpe	10.7%	20.1%	0.61	-36.0%	1.67x
Equity Minimum-Variance	7.0%	12.7%	0.60	-17.8%	0.70x
Equity Risk Parity	10.0%	14.9%	0.71	-19.4%	0.64x
Combined Equal-Weight	15.0%	21.6%	0.76	-27.9%	0.79x
Combined Maximum-Sharpe	7.1%	32.7%	0.37	-48.6%	2.08x
Combined Minimum-Variance	7.0%	12.7%	0.60	-17.8%	0.70x
Combined Risk Parity	12.7%	17.4%	0.77	-22.4%	0.71x
Crypto Equal-Weight	40.4%	81.9%	0.83	-81.6%	1.03x
Crypto Maximum-Sharpe	40.0%	77.0%	0.82	-78.0%	1.28x
Crypto Minimum-Variance	53.6%	74.3%	0.95	-72.2%	1.13x
Crypto Risk Parity	41.6%	80.3%	0.84	-80.5%	1.03x

Exhibit A1. Complete net performance metrics table.

Source: results/tables/performance_metrics.csv. Return is compound annual growth; volatility and Sharpe use 252 for equity/combined and 365 for crypto; zero risk-free rate.

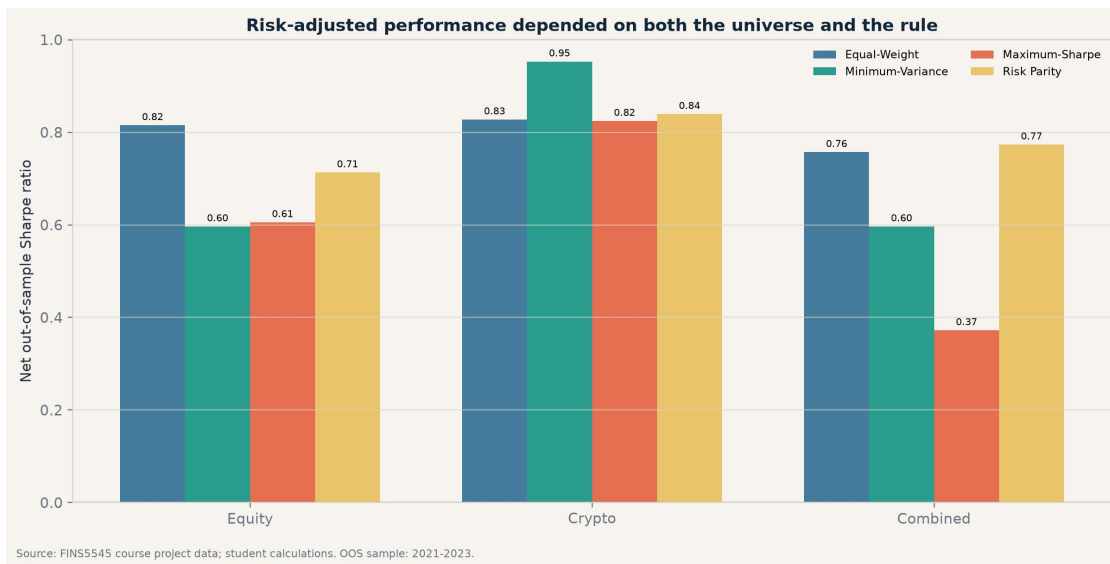


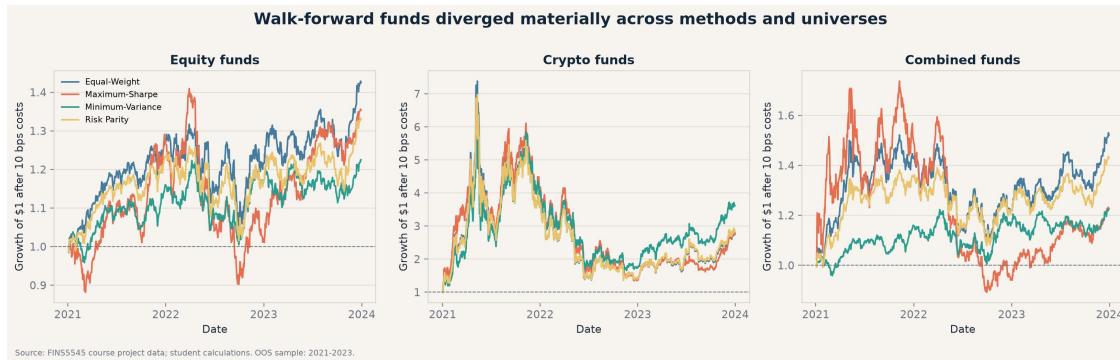
Exhibit A2. Net out-of-sample Sharpe ratio by method and family.

Source: FIN5545 prices; student calculations; 2021–2023; 10 bps one-way cost.

Crypto funds lead the sample Sharpe rankings but operate at a much higher risk level. Within equities, the simple Equal-Weight rule remains difficult to improve upon. Across the combined universe, Risk Parity only narrowly exceeds Equal-Weight on Sharpe ratio.

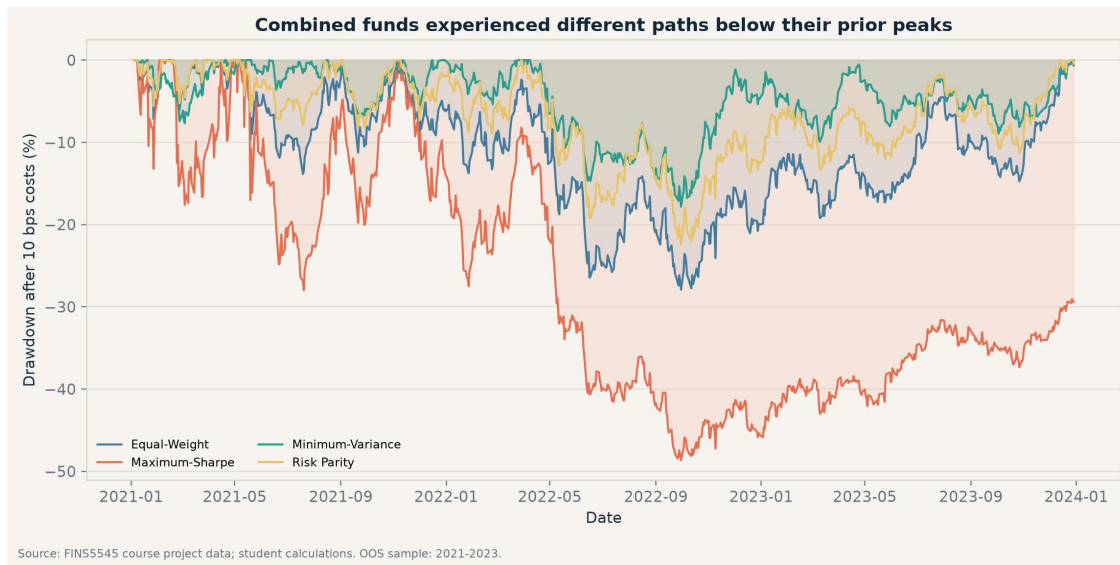
APPENDIX B

Required exhibits: growth and drawdown

**Exhibit B1. Growth of \$1 across all twelve funds, grouped by universe.**

Source: FIN5545 prices; student calculations; net of 10 bps one-way costs; 2021–2023.

Crypto growth dominates the chart's vertical scale but is accompanied by much larger reversals. Separate equity and combined panels are therefore more informative for users with moderate risk tolerance.

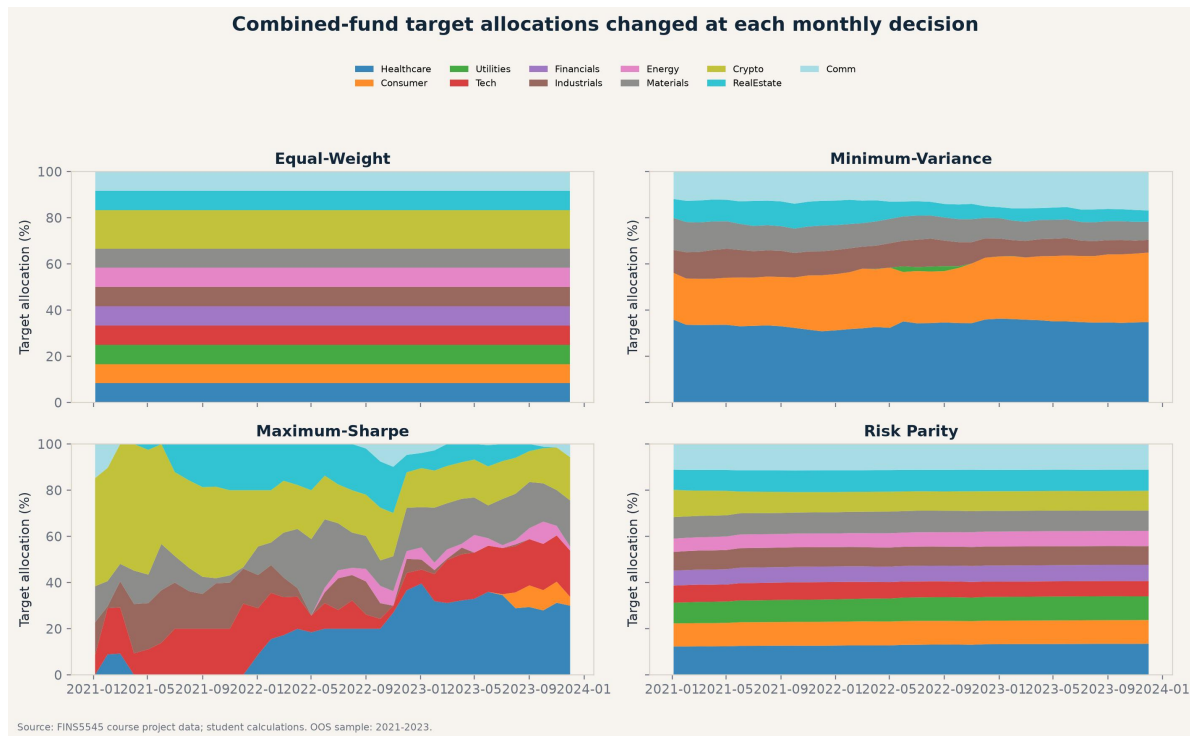
**Exhibit B2. Drawdown paths for the four combined funds.**

Source: FIN5545 prices; student calculations; net of 10 bps one-way costs; 2021–2023.

In the combined universe, Risk Parity has a smaller worst drawdown than Equal-Weight, while Maximum-Sharpe is considerably less stable.

APPENDIX C

Required exhibit: portfolio weights over time

**Exhibit C1. Monthly target allocations by group for all four combined-fund methods.**

Source: FINSS545 prices; expanding estimates use prior observations only; 2021–2023.

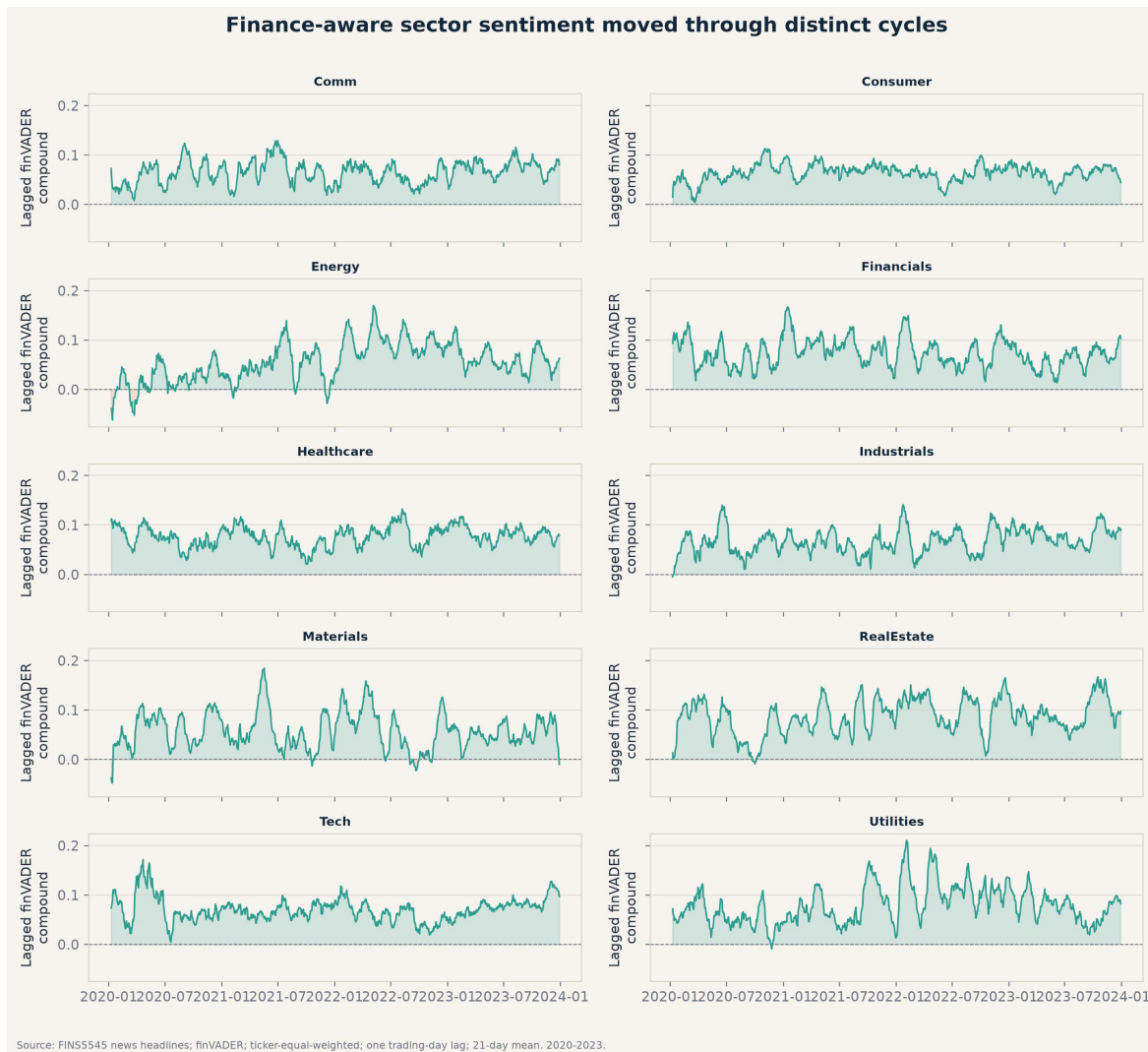
Equal-Weight does not change by construction. Minimum-Variance largely avoids crypto, Risk Parity adjusts allocations gradually, and Maximum-Sharpe makes pronounced shifts as estimated returns change.

Audit link

The underlying ticker-level monthly targets are in `results/data/fund_weights.csv`. `results/tables/portfolio_constraint_checks.csv` records weight sum, minimum, and maximum for every fund-rebalance pair.

APPENDIX D

Required exhibit: sector sentiment over time

**Exhibit D1. Lagged 21-day finVADER sentiment by sector, 2020–2023.**

Source: FINSS545 headlines; headline scores average within ticker-day, then equally across tickers; five-day carry; one trading-day lag.

Sector sentiment follows broad market conditions but still shows meaningful differences across sectors. Equal-ticker aggregation prevents one company from having too much influence simply because it receives more news coverage.

Interpretation

The sector chart is a descriptive state variable. It is not itself a trading result. Any allocation use is evaluated separately in Appendix E with the prescribed lag, holdout, turnover, and costs.

APPENDIX E

Sentiment diagnostics and fear/greed

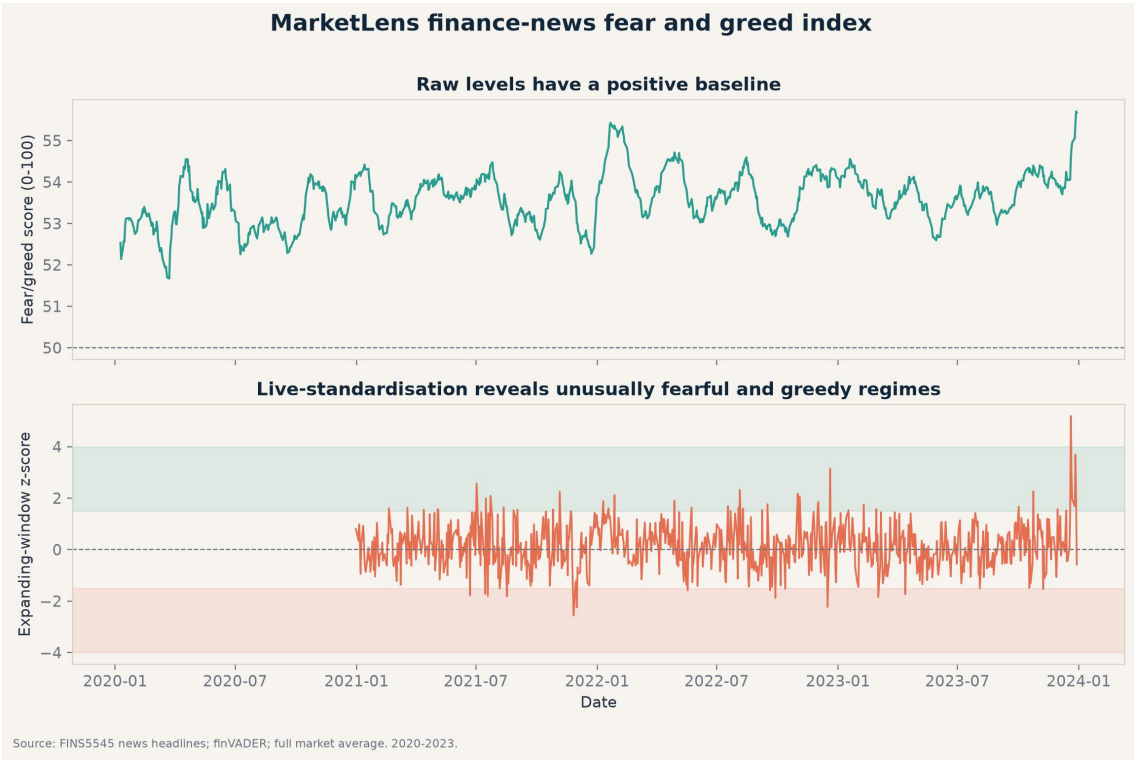


Exhibit E1. MarketLens news fear/greed level and expanding live z-score.

Source: FIN5545 headlines; finVADER; equal headline contribution after ticker-day aggregation; 2020–2023.

The live z-score uses only information available by each date. A full-sample standardisation is retained in the CSV for descriptive comparison, but it plays no role in live decisions.

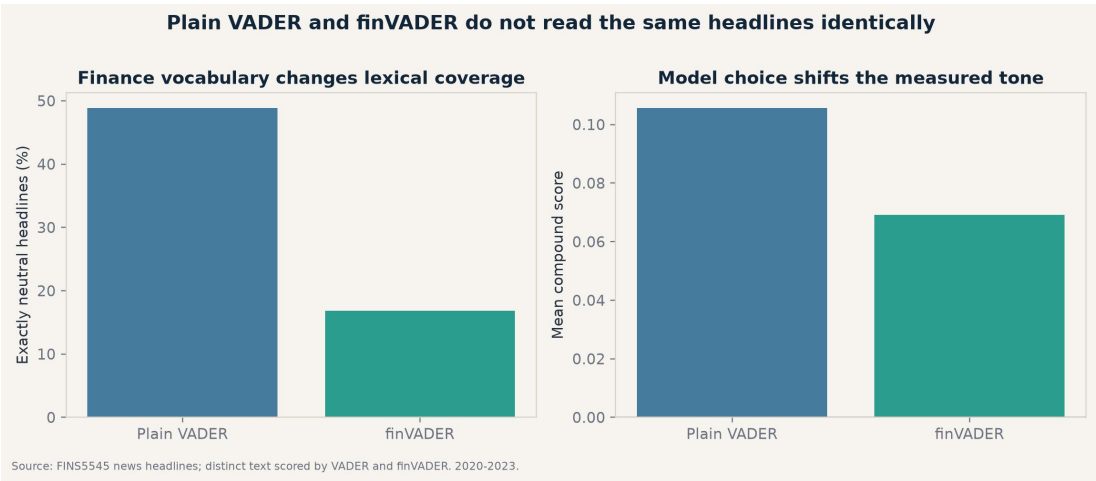


Exhibit E2. Sentiment model coverage robustness.

Source: results/tables/sentiment_model_validation.csv; 2020–2023.

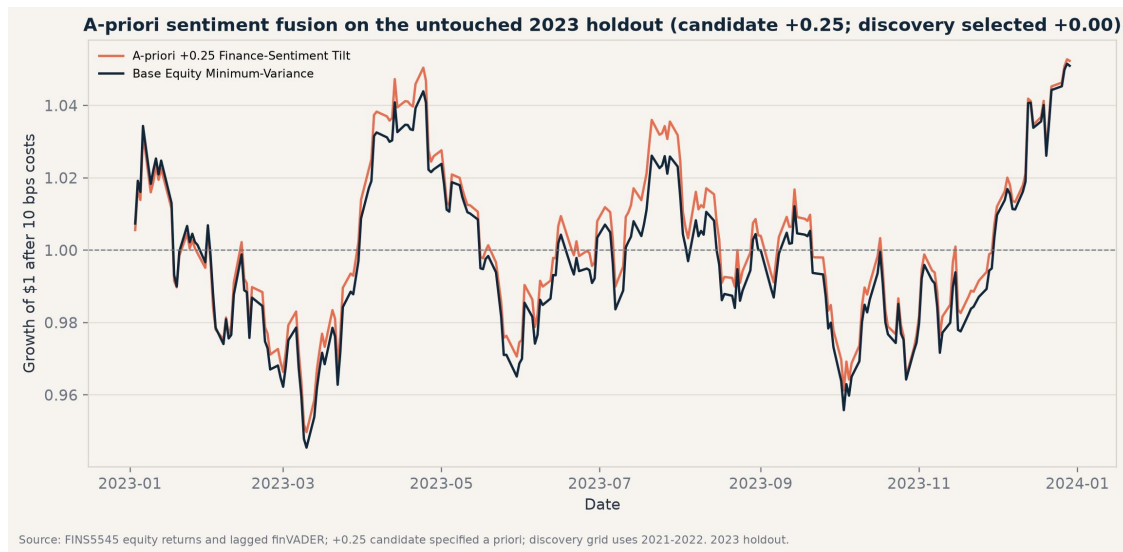
APPENDIX F

Required fusion evidence and robustness

Period	Model	Return	Sharpe	MDD	Turnover
Full OOS (descriptive; strength tuned within period)	Base Equity Minimum-Variance	7.0%	0.60	-17.8%	2.09x
Full OOS (descriptive; strength tuned within period)	Finance-Sentiment Tilt	6.9%	0.59	-18.1%	4.79x
Discovery (strength selected here)	Base Equity Minimum-Variance	8.0%	0.63	-17.8%	1.82x
Discovery (strength selected here)	Finance-Sentiment Tilt	7.7%	0.62	-18.1%	3.75x
Untouched holdout	Base Equity Minimum-Variance	5.1%	0.51	-8.6%	0.27x
Untouched holdout	Finance-Sentiment Tilt	5.3%	0.52	-8.5%	1.04x

Exhibit F1. Base versus pre-set +0.25 sentiment candidate in discovery and untouched holdout periods.

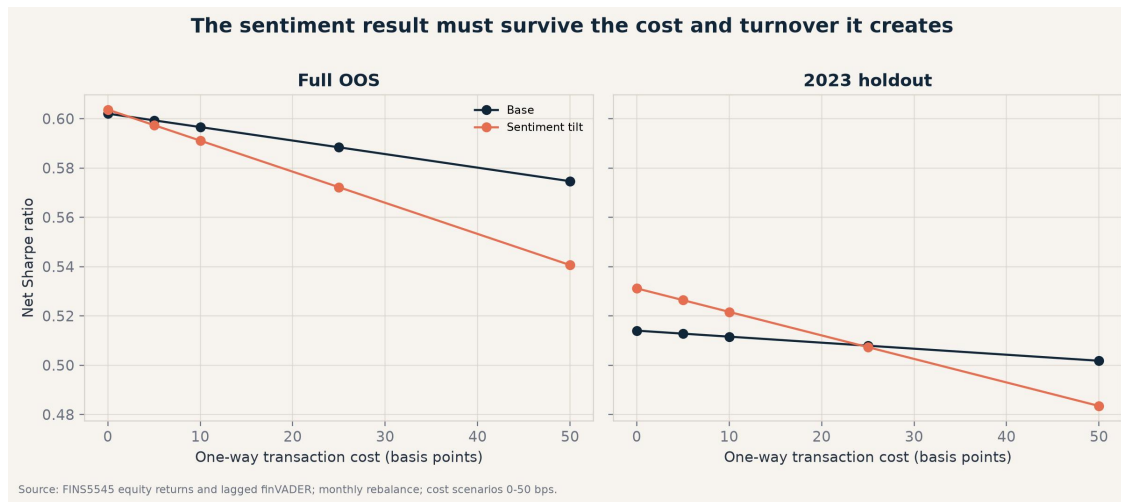
Source: results/tables/fusion_performance.csv; net of 10 bps one-way costs.

**Exhibit F2. Required before/after fusion growth comparison in 2023.**

Source: FIN5545 equity returns and lagged finVADER; student calculations.

APPENDIX F (CONTINUED)

Transaction-cost robustness

**Exhibit F3. Fusion Sharpe sensitivity to one-way trading costs and turnover.**

Source: results/tables/fusion_cost_robustness.csv; cost scenarios 0–50 bps.

The candidate's advantage is not robust. It disappears at about 25 bps in the holdout and is weaker across the full out-of-sample period. For now, the tilt should remain a research result rather than a product feature.

APPENDIX G

Reproducibility, limitations, and references

Reproducibility chain

Layer	Control
Data	Official hosted ZIP loaded only through <code>src/data_access.py</code> ; raw files excluded.
Build	<code>scripts/run_part_b.py</code> recreates cleaned panels, all funds, sentiment, fusion, tables, and figures.
Deploy	<code>streamlit_app.py</code> loads only committed results CSVs; no runtime data download, NLP, or optimisation.
Tests	pytest checks future-data invariance, required artifacts, weight sums, non-negativity, and known metrics; Streamlit AppTest reports zero exceptions.
AI audit	AGENTS.md and ai/ record the constraints, prompts, risks, rejected selection, chart corrections, and student responsibilities.

Exhibit G1. Reproduction and hand-in controls.

Source: project repository; see README.md and docs/FINAL_HANDOFF.md.

Limitations

The 2021–2023 out-of-sample period is short and may cover only a limited range of market regimes. Headlines lack the context of full articles, and lexical sentiment may misread sarcasm, negation, or entity scope. The news data cover equities rather than crypto, while optimisation remains sensitive to noisy inputs. Transaction costs are scenario assumptions rather than broker-specific estimates. Taxes, slippage, market impact, and operating costs are outside the project scope.

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